

DO-IT-YOURSELF: PROJECT

supporting components. The output voltage and current can be configured using the solder jumpers on D0 and D1, as shown in the configuration table discussed earlier. In the shown setup, a 3.6V output is used, with both D0 and D1 set to state '1'.

The next step is to design a supercapacitor charging and storage system capable of harvesting energy from multiple sources. One source is a piezoelectric transducer for harvesting vibration and impact energy, while the other is a solar cell for harvesting ambient light. The collected energy is stored in supercapacitors, which can be connected to low-power ICs or to a battery-charging circuit for charging small batteries, making the system suitable for IoT and other low-power applications.

In this design, the LTC3588 module serves as the primary energy-harvesting interface. The piezo input connects through the PZ1 and PZ2 pins, and the supercapacitor stores harvested energy for later use with an MCU or other low-power electronics.

A solar cell may also be included to charge the supercapacitor alongside the piezo source, ensuring continuous charging whenever ambient light is available. The output voltage and current can be configured using the D0 and D1 pins. The configuration details are available in Table 2.

Construction and testing of LTC3588-based circuit. This design includes a supercapacitor charging and storage system capable of harvesting energy from multiple sources. A piezoelectric element captures vibration and impact energy, while a solar cell harvests ambient light. Harvested energy is stored in supercapacitors, which may then be used to charge small batteries or power low-power IoT devices.

The LTC3588 module manages energy extraction, with the piezo input connected through PZ1 and PZ2. The supercapacitor stores and supplies a charge to an MCU board or similar electronics. A solar cell may be incorporated to charge the supercapacitor under ambient light, ensuring continuous charging. The output voltage and current are configurable using the D0 and D1 pins as shown in Table 2.

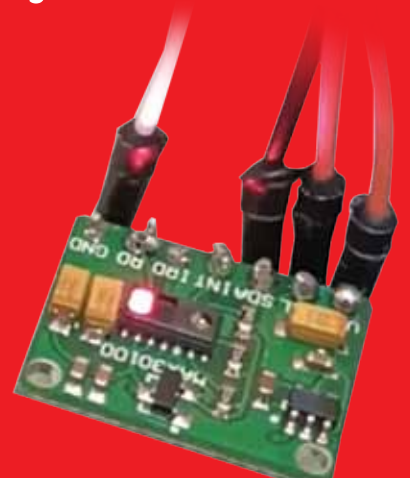
For testing, vibration is applied to the piezoelectric element connected to the system. Light tapping with a small hammer enables the element to harvest energy and charge the connected supercapacitor. When an LED is connected between VOUT and GND, it illuminates fully, demonstrating battery-less operation driven solely by harvested energy. **EFY**

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